A SIX TERM EXACT SEQUENCE FOR THE NAIVE COTANGENT COMPLEX OF A COMPOSITION OF RING MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The cone of a map of complexes and the criterion for producing a map into a shift of a cone from a map plus an explicit homotopy. Both are invoked inside the counted proof (source lines 6034-6043) as cross-chapter pointers. Formulate the canonical comparison map that the statement requires.
- The Description of the naive cotangent complex through stalks, and the algebraic exact sequence for a composition of ring maps. Invoked near the end of the counted proof (source lines 6055-6059) as pointers to a lemma of this chapter and to a result of another chapter. Reduce the exactness statement to the corresponding algebraic statement.

Source reference. Original reference derived-definition-cone: sources/stacks-modules/pinned-original-source.tex, source lines 6007–6007. Navigation pointer only: the counted text refers to the Stacks item derived-definition-cone, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference derived-lemma-map-from-cone: sources/stacks-modules/pinned-original-source.tex, source lines 6043–6043. Navigation pointer only: the counted text refers to the Stacks item derived-lemma-map-from-cone, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-stalk-NL: sources/stacks-modules/pinned-original-source.tex, source lines 6057–6057. Navigation pointer only: the counted

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Source reference. Original reference algebra-lemma-exact-sequence-NL: sources/stacks-modules/pinned-original-source.tex, source lines 6059–6059. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-exact-sequence-NL, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-remark-composition-homotopy-equivalent-to-zero: sources/stacks-modules/pinned-original-source.tex, source lines 6054–6054. Navigation pointer only: the counted text refers to the Stacks item algebra-remark-composition-homotopy-equivalent-to-zero, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. Let $\mathcal{A} \to \mathcal{B} \to \mathcal{C}$ be maps of sheaves of rings. Let C be the cone (Derived Categories, Definition derived-definition-cone) of the map of complexes $NL_{\mathcal{C}/\mathcal{A}} \to NL_{\mathcal{C}/\mathcal{B}}$. There is a canonical map*

$$c : NL_{\mathcal{B}/\mathcal{A}} \otimes_{\mathcal{B}} \mathcal{C} \longrightarrow C[-1]$$

of complexes of $\mathcal{C}$-modules which produces a canonical six term exact sequence

$$H^0(NL_{\mathcal{B}/\mathcal{A}} \otimes_{\mathcal{B}} \mathcal{C}) \longrightarrow H^0(NL_{\mathcal{C}/\mathcal{A}}) \longrightarrow H^0(NL_{\mathcal{C}/\mathcal{B}}) \longrightarrow 0$$

$$H^{-1}(NL_{\mathcal{B}/\mathcal{A}} \otimes_{\mathcal{B}} \mathcal{C}) \longrightarrow H^{-1}(NL_{\mathcal{C}/\mathcal{A}}) \longrightarrow H^{-1}(NL_{\mathcal{C}/\mathcal{B}})$$

of cohomology sheaves.

Proof. To give the map c we have to give a map $c_1 : NL_{\mathcal{B}/\mathcal{A}} \otimes_{\mathcal{B}} \mathcal{C} \to NL_{\mathcal{C}/\mathcal{A}}$ and an explicit homotopy between the composition

$$NL_{\mathcal{B}/\mathcal{A}} \otimes_{\mathcal{B}} \mathcal{C} \to NL_{\mathcal{C}/\mathcal{A}} \to NL_{\mathcal{C}/\mathcal{B}}$$

and the zero map, see Derived Categories, Lemma derived-lemma-map-from-cone. For c_1 we use the functoriality described above for the obvious diagram. For the homotopy we use the map

$$NL^0_{\mathcal{B}/\mathcal{A}} \otimes_{\mathcal{B}} \mathcal{C} \longrightarrow NL^{-1}_{\mathcal{C}/\mathcal{B}}, \quad \mathrm{d}[b] \otimes 1 \longmapsto [\varphi(b)] - b[1]$$

where $\varphi : \mathcal{B} \to \mathcal{C}$ is the given map. Please compare with Algebra, Remark algebra-remark-composition-homotopy-equivalent-to-zero. To see the consequence for cohomology sheaves, it suffices to show that $H^0(c)$ is an isomorphism and $H^{-1}(c)$ surjective. To see this we can look at stalks, see Lemma lemma-stalk-NL, and then we can use the corresponding result in commutative algebra, see Algebra, Lemma algebra-lemma-exact-sequence-NL. Some details omitted. □